

Logistic Regression

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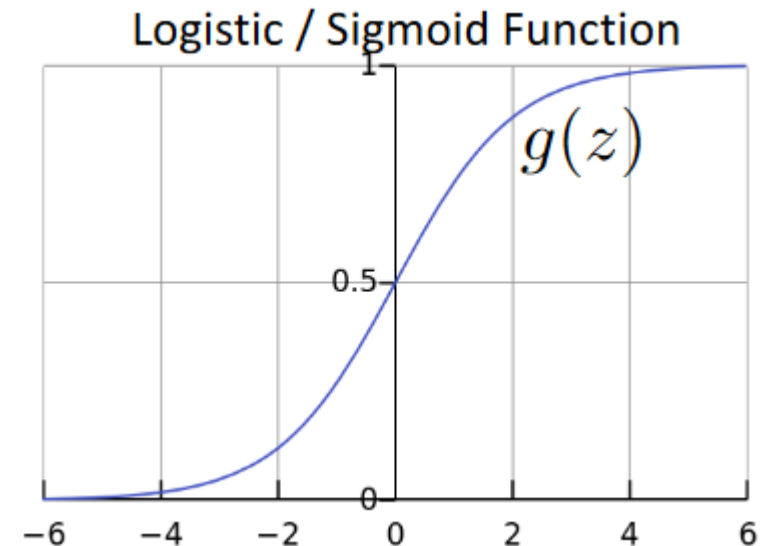
- It takes a probabilistic approach to learn a classifier (a function)
- $h_{\theta}(\mathbf{x})$ should give $p(y = 1|\mathbf{x}; \theta)$
- We want $0 \leq h_{\theta}(\mathbf{x}) \leq 1$
- Logistic regression model :

$$h_{\theta}(\mathbf{x}) = g(\theta^T \mathbf{x})$$

$$g(z) = \frac{1}{1 + e^{-z}}$$

$$h_{\theta}(\mathbf{x}) = \frac{1}{1 + e^{-\theta^T \mathbf{x}}} = \frac{e^{\theta^T \mathbf{x}}}{1 + e^{\theta^T \mathbf{x}}}$$

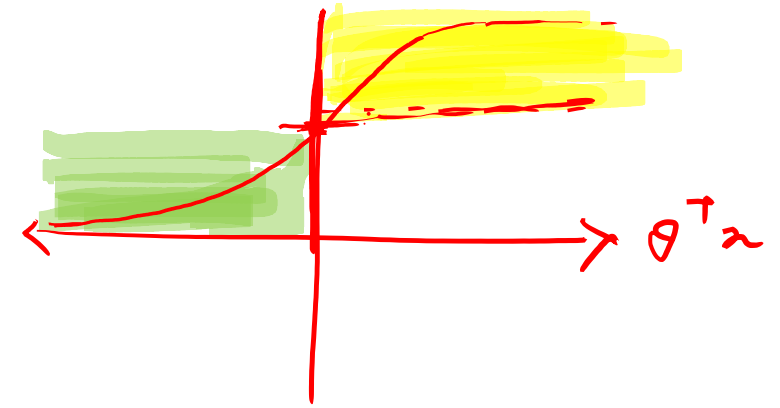
- The sigmoid first computes real-valued score and then squashes it between 0 and 1 to make it as a probability score.



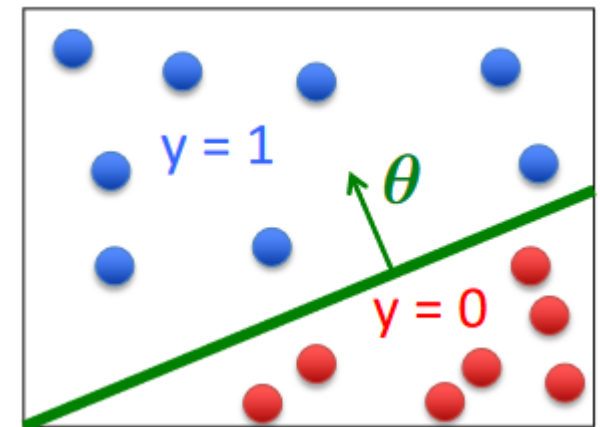
Interpreting Hypothesis Output

- $h_{\boldsymbol{\theta}}(\mathbf{x}) = \text{estimated } p(y = 1|\mathbf{x}; \boldsymbol{\theta}) = \frac{1}{1+e^{-\boldsymbol{\theta}^T \mathbf{x}}} = \frac{e^{\boldsymbol{\theta}^T \mathbf{x}}}{1+e^{\boldsymbol{\theta}^T \mathbf{x}}}$
- Note: $p(y = 0|\mathbf{x}; \boldsymbol{\theta}) + p(y = 1|\mathbf{x}; \boldsymbol{\theta}) = 1$
- So, $p(y = 0|\mathbf{x}; \boldsymbol{\theta}) = 1 - p(y = 1|\mathbf{x}; \boldsymbol{\theta}) = 1 - \frac{1}{1+e^{-\boldsymbol{\theta}^T \mathbf{x}}} = \frac{1}{1+e^{\boldsymbol{\theta}^T \mathbf{x}}}$
- The *log-odds (logits)* of the model
$$\log \frac{p(y = 1|\mathbf{x}; \boldsymbol{\theta})}{p(y = 0|\mathbf{x}; \boldsymbol{\theta})} = \log e^{\boldsymbol{\theta}^T \mathbf{x}} = \boldsymbol{\theta}^T \mathbf{x}$$
- Thus if $\boldsymbol{\theta}^T \mathbf{x} > 0$ then the positive class more probable

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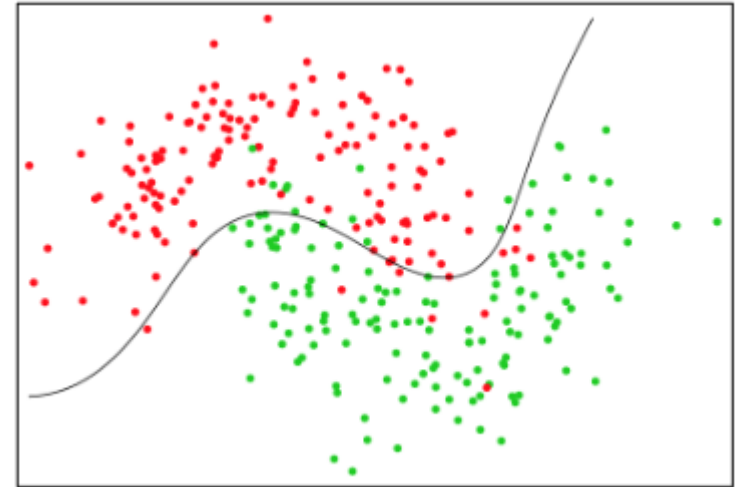
- $h_{\theta}(\mathbf{x}) = g(\theta^T \mathbf{x})$
- $g(z) = \frac{1}{1+e^{-z}}$
- $\theta^T \mathbf{x}$ should be large negative values for negative instances
- $\theta^T \mathbf{x}$ should be large positive values for positive instances
- Assume a threshold and predict
 - $y = 1$ if $h_{\theta}(\mathbf{x}) \geq 0.5$ ($\theta^T \mathbf{x} \geq 0$)
 - $y = 0$ if $h_{\theta}(\mathbf{x}) < 0.5$ ($\theta^T \mathbf{x} < 0$)



Non-linear Decision Boundary

- Can apply basis function expansion to features

$$\bullet \mathbf{x} = \begin{bmatrix} 1 \\ x_1 \\ x_2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ x_1 x_2 \\ x_1^2 \\ x_2^2 \\ x_1^2 x_2 \\ x_1 x_2^2 \\ \vdots \end{bmatrix}$$



$$h_{\theta}(x) = g(\theta^T x)$$

Let $\theta = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$ ↓ bias



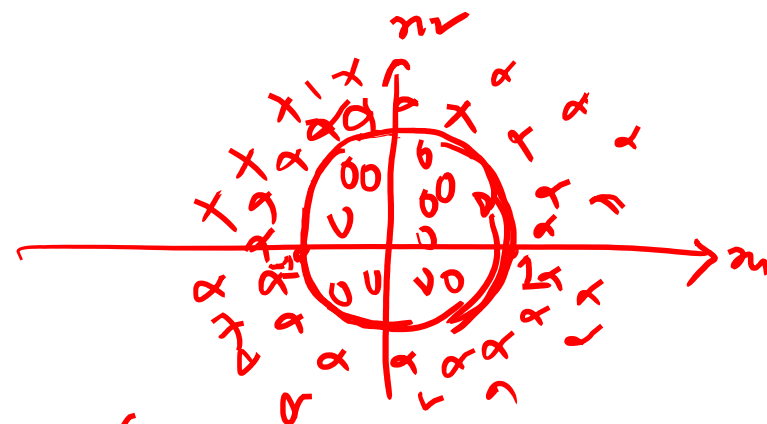
$$h_{\theta}(x) = (-3 + x_1 + x_2)$$

$$\theta^T x = 0$$

$$x_1 + x_2 = 3$$

$$\text{If } x_1 + x_2 > 3$$

$$\theta = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2)$$

$$\left. \begin{array}{l} \theta^T \phi(x) \geq 0 \\ x_1^2 + x_2^2 \geq 1 \end{array} \right\} y = 1$$

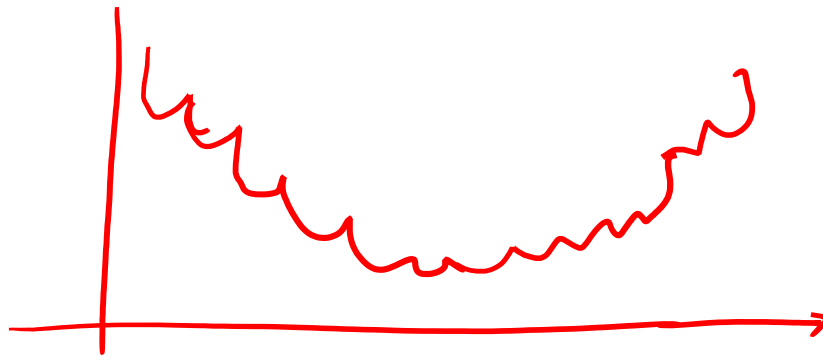
Logistic Regression Cost Function

- Should not use the squared loss as in case of linear regression

$$J(\boldsymbol{\theta}) = \frac{1}{2n} \sum_{i=1}^n (h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) - y^{(i)})^2$$

- The logistic regression model will lead to a non-convex cost function

$$h_{\boldsymbol{\theta}}(\mathbf{x}) = \frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{x}}}$$



Finding the Cost Function via MLE

- Likelihood of the data is given by $L(\boldsymbol{\theta}) = \prod_{i=1}^n p(y^{(i)} | \mathbf{x}^{(i)}; \boldsymbol{\theta})$
- $\boldsymbol{\theta}$ that maximizes the likelihood

$$\boldsymbol{\theta}_{MLE} = \arg \max_{\boldsymbol{\theta}} L(\boldsymbol{\theta}) = \arg \max_{\boldsymbol{\theta}} \prod_{i=1}^n p(y^{(i)} | \mathbf{x}^{(i)}; \boldsymbol{\theta})$$

$$\begin{aligned} \boldsymbol{\theta}_{MLE} &= \arg \max_{\boldsymbol{\theta}} \log L(\boldsymbol{\theta}) = \arg \max_{\boldsymbol{\theta}} \log \prod_{i=1}^n p(y^{(i)} | \mathbf{x}^{(i)}; \boldsymbol{\theta}) \\ &= \arg \max_{\boldsymbol{\theta}} \sum_{i=1}^n \log p(y^{(i)} | \mathbf{x}^{(i)}; \boldsymbol{\theta}) \end{aligned}$$

Finding the Cost Function via MLE

- Each label $y^{(i)}$ is binary with probability $h_{\theta}(\mathbf{x}^{(i)})$
- Assume Bernoulli likelihood

$$\begin{aligned} p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) &= \prod_{i=1}^n p(y^{(i)} | \mathbf{x}^{(i)}; \boldsymbol{\theta}) \\ &= \prod_{i=1}^n h_{\theta}(\mathbf{x}^{(i)})^{y^{(i)}} \left(1 - h_{\theta}(\mathbf{x}^{(i)})\right)^{1-y^{(i)}} \end{aligned}$$

- The log-likelihood

$$l(\boldsymbol{\theta}) = \sum_{i=1}^n y^{(i)} \log h_{\theta}(\mathbf{x}^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(\mathbf{x}^{(i)}))$$

The Cost Function

- Maximizing $l(\boldsymbol{\theta})$ is equivalent to minimizing the NLL

$$J(\boldsymbol{\theta}) = NLL(\boldsymbol{\theta}) = - \sum_{i=1}^n y^{(i)} \log h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}))$$

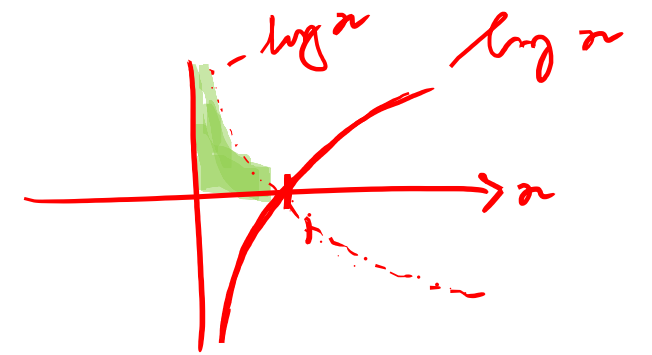
- Cost of a single instance

$$cost(h_{\boldsymbol{\theta}}(\mathbf{x}), y) = \begin{cases} -\log h_{\boldsymbol{\theta}}(\mathbf{x}) & \text{if } y = 1 \\ -\log (1 - h_{\boldsymbol{\theta}}(\mathbf{x})) & \text{if } y = 0 \end{cases}$$

- The objective function

$$J(\boldsymbol{\theta}) = \sum_{i=1}^n cost(h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}), y^{(i)})$$

Intuition



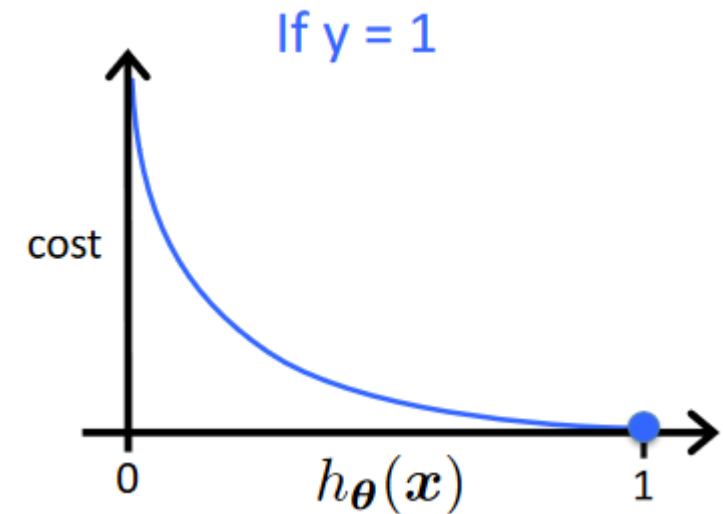
$$\bullet \text{ cost}(h_{\theta}(\mathbf{x}), y) = \begin{cases} -\log h_{\theta}(\mathbf{x}) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(\mathbf{x})) & \text{if } y = 0 \end{cases}$$

if $y = 1$

if $y = 0$

- If $y = 1$

- $\text{cost} = 0$ for correct prediction
- As $h_{\theta}(\mathbf{x}) \rightarrow 0$, $\text{cost} \rightarrow \infty$
- Mistakes should get large penalties
- e.g., predict $h_{\theta}(\mathbf{x}) = 0$, but $y = 1$

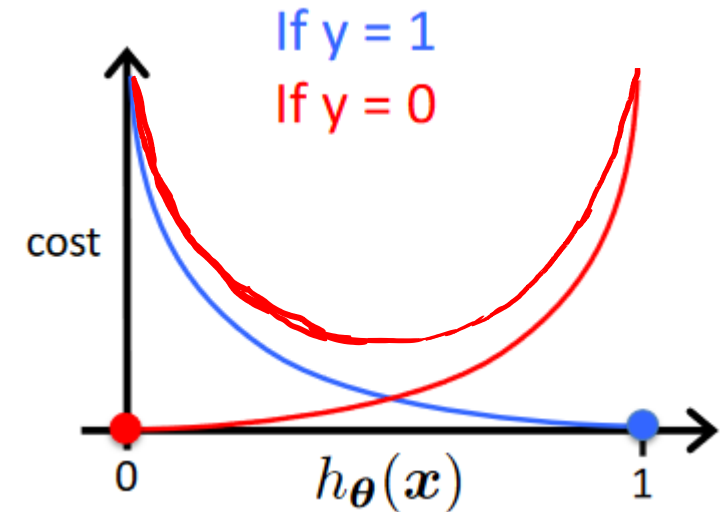


Intuition

- $cost(h_{\theta}(x), y) = \begin{cases} -\log h_{\theta}(x) \\ -\log (1 - h_{\theta}(x)) \end{cases}$

if $y = 1$
if $y = 0$

- If $y = 0$
 - $cost = 0$ for correct prediction
 - As $(1 - h_{\theta}(x)) \rightarrow 0$, $cost \rightarrow \infty$
 - Mistakes should get large penalties
 - e.g., predict $h_{\theta}(x) = 0$, but $y = 1$



Logistic Regression Cost fn. $z^{(i)} = \theta^T x^{(i)}$; $\sigma(z) = \frac{1}{1 + e^{-z}}$

$$J(\theta) = - \sum_{i=1}^n \left[y^{(i)} \log \sigma(z^{(i)}) + (1 - y^{(i)}) \log (1 - \sigma(z^{(i)})) \right]$$

$$= - \sum_{i=1}^n \left[\underbrace{y^{(i)} \log \frac{1}{1 + e^{-z^{(i)}}}}_{\text{1st term}} + \underbrace{(1 - y^{(i)}) \log \frac{e^{-z^{(i)}}}{1 + e^{-z^{(i)}}}}_{\text{2nd term}} \right]$$

1st Term:- $y^{(i)} \log \frac{1}{1 + e^{-z^{(i)}}} = - y^{(i)} \log (1 + e^{-z^{(i)}})$

2nd term:- $\log \frac{e^{-z}}{1 + e^{-z}} = \log e^{-z} - \log (1 + e^{-z})$

$$= -z - \log (1 + e^{-z})$$

$$(1 - y^{(i)}) \log \frac{e^{-z}}{1 + e^{-z}} = - (1 - y^{(i)}) z^{(i)} - (1 - y^{(i)}) \log (1 + e^{-z^{(i)}})$$

$$J(\theta) = \underbrace{- y^{(i)} \log (1 + e^{-z^{(i)}})}_{\text{1st term}} - \underbrace{(1 - y^{(i)}) z^{(i)} - (1 - y^{(i)}) \log (1 + e^{-z^{(i)}})}_{\text{2nd term}}$$

$$= - \sum_{i=1}^n \left[-\log(1 + e^{-z^{(i)}}) - (1 - y^{(i)}) z^{(i)} \right]$$

$$= \sum_{i=1}^n \left[\log(1 + e^{-z^{(i)}}) + (1 - y^{(i)}) z^{(i)} \right]$$

$$= \sum_{i=1}^n \left[\log(1 + e^{z^{(i)}}) - \log e^{z^{(i)}} + z^{(i)} - y^{(i)} z^{(i)} \right] = \frac{1 + e^z}{e^z}$$

$$J = \sum_{i=1}^n \left[\log(1 + e^{z^{(i)}}) - y^{(i)} z^{(i)} \right]$$

$$\frac{\partial J}{\partial \theta} = \frac{\partial}{\partial \theta} \left[\sum_{i=1}^n \left(\log(1 + e^{\theta^T x^{(i)}}) - y^{(i)} \theta^T x^{(i)} \right) \right]$$

$$= \sum_{i=1}^n \left(\frac{e^{\theta^T x^{(i)}}}{1 + e^{\theta^T x^{(i)}}} \cdot x^{(i)} - y^{(i)} x^{(i)} \right)$$

$$\begin{aligned}\frac{\partial J}{\partial \theta} &= \sum_{i=1}^n \left(\lambda_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)} \\ &= X^T \left(\lambda_{\theta}(\vec{x}^{(1)}) - \vec{y} \right) \longrightarrow 0\end{aligned}$$

we cannot get a closed form solution for $\vec{\theta}$
by $\frac{\partial J}{\partial \theta} = 0$.